

Chapter 4: Organic Chemistry & Compounds Syllabus

Class 10 Chemistry • Official Course Syllabus & Practice Sheet

1. Syllabus Breakdown & Chapter Overview

This syllabus covers the basics of Carbon compounds, functional groups, and structural formulas in organic chemistry.

- **Hydrocarbons:** Alkanes (single bonds, saturated), Alkenes (double bonds, unsaturated), and Alkynes (triple bonds, unsaturated).
- **Functional Groups:** Study of Halogens, Alcohols (-OH), Aldehydes (-CHO), Ketones (C=O), and Carboxylic acids (-COOH).
- **Isomerism:** Organic compounds sharing the same molecular formula but exhibiting different structural configurations.
- **Chemical Properties:** Combustion, oxidation, addition reactions, and substitution reactions of hydrocarbons.

2. Board Exam Reference Questions

Q1. [Short Answer - 3 Marks]

Explain why covalent bonding is highly characteristic of carbon atoms. Discuss catenation and tetravalency.

Q2. [Long Answer - 7 Marks]

Define homologous series. State any four characteristics of a homologous series. Write the IUPAC name and draw the structure of Ethanol and Ethanoic Acid.

3. Revision Checklist

- Memorize the general formulas for Alkanes ($C_n H_{2n+2}$), Alkenes ($C_n H_{2n}$), and Alkynes ($C_n H_{2n-2}$).
 - Practice drawing Lewis dot structures for Methane, Ethane, and Ethene.
 - Understand the saponification reaction and the cleansing action of soaps and detergents.
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